

Descriptive Statistics: Solve the following problems

16. A bowler's scores for six games were 182, 168, 184, 190, 170, and 174. Using these data as a sample, compute the following descriptive statistics:
- Range
 - Variance
 - Standard deviation
 - Coefficient of variation
17. A home theater in a box is the easiest and cheapest way to provide surround sound for a home entertainment center. A sample of prices is shown here (*Consumer Reports Buying Guide*, 2004). The prices are for models with a DVD player and for models without a DVD player.

Models with DVD Player	Price	Models without DVD Player	Price
Sony HT-1800DP	\$450	Pioneer HTP-230	\$300
Pioneer HTD-330DV	300	Sony HT-DDW750	300
Sony HT-C800DP	400	Kenwood HTB-306	360
Panasonic SC-HT900	500	RCA RT-2600	290
Panasonic SC-MTI	400	Kenwood HTB-206	300

- Compute the mean price for models with a DVD player and the mean price for models without a DVD player. What is the additional price paid to have a DVD player included in a home theater unit?
 - Compute the range, variance, and standard deviation for the two samples. What does this information tell you about the prices for models with and without a DVD player?
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18. Car rental rates per day for a sample of seven Eastern U.S. cities are as follows (*The Wall Street Journal*, January 16, 2004).

City	Daily Rate
Boston	\$43
Atlanta	35
Miami	34
New York	58
Orlando	30
Pittsburgh	30
Washington, D.C.	36

- Compute the mean, variance, and standard deviation for the car rental rates.
 - A similar sample of seven Western U.S. cities showed a sample mean car rental rate of \$38 per day. The variance and standard deviation were 12.3 and 3.5, respectively. Discuss any difference between the car rental rates in Eastern and Western U.S. cities.
19. The *Los Angeles Times* regularly reports the air quality index for various areas of Southern California. A sample of air quality index values for Pomona provided the following data: 28, 42, 58, 48, 45, 55, 60, 49, and 50.
- Compute the range and interquartile range.
 - Compute the sample variance and sample standard deviation.
 - A sample of air quality index readings for Anaheim provided a sample mean of 48.5, a sample variance of 136, and a sample standard deviation of 11.66. What comparisons can you make between the air quality in Pomona and that in Anaheim on the basis of these descriptive statistics?

20. The following data were used to construct the histograms of the number of days required to fill orders for Dawson Supply, Inc., and J.C. Clark Distributors (see Figure 3.2).

Dawson Supply Days for Delivery: 11 10 9 10 11 11 10 11 10 10
Clark Distributors Days for Delivery: 8 10 13 7 10 11 10 7 15 12

Use the range and standard deviation to support the previous observation that Dawson Supply provides the more consistent and reliable delivery times.

21. How do grocery costs compare across the country? Using a market basket of 10 items including meat, milk, bread, eggs, coffee, potatoes, cereal, and orange juice, *Where to Retire* magazine calculated the cost of the market basket in six cities and in six retirement areas across the country (*Where to Retire*, November/December 2003). The data with market basket cost to the nearest dollar are as follows:

City	Cost	Retirement Area	Cost
Buffalo, NY	\$33	Biloxi-Gulfport, MS	\$29
Des Moines, IA	27	Asheville, NC	32
Hartford, CT	32	Flagstaff, AZ	32
Los Angeles, CA	38	Hilton Head, SC	34
Miami, FL	36	Fort Myers, FL	34
Pittsburgh, PA	32	Santa Fe, NM	31

- Compute the mean, variance, and standard deviation for the sample of cities and the sample of retirement areas.
 - What observations can be made based on the two samples?
23. Scores turned in by an amateur golfer at the Bonita Fairways Golf Course in Bonita Springs, Florida, during 2005 and 2006 are as follows:
- 2005 Season:* 74 78 79 77 75 73 75 77
2006 Season: 71 70 75 77 85 80 71 79
- Use the mean and standard deviation to evaluate the golfer's performance over the two-year period.
 - What is the primary difference in performance between 2005 and 2006? What improvement, if any, can be seen in the 2006 scores?
24. The following times were recorded by the quarter-mile and mile runners of a university track team (times are in minutes).

Quarter-Mile Times: .92 .98 1.04 .90 .99
Mile Times: 4.52 4.35 4.60 4.70 4.50

After viewing this sample of running times, one of the coaches commented that the quarter-milers turned in the more consistent times. Use the standard deviation and the coefficient of variation to summarize the variability in the data. Does the use of the coefficient of variation indicate that the coach's statement should be qualified?

33. Florida Power & Light (FP&L) Company has enjoyed a reputation for quickly fixing its electric system after storms. However, during the hurricane seasons of 2004 and 2005, a new reality was that the company's historical approach to emergency electric system repairs was no longer good enough (*The Wall Street Journal*, January 16, 2006). Data showing the days required to restore electric service after seven hurricanes during 2004 and 2005 follow.

Hurricane	Days to Restore Service
Charley	13
Frances	12
Jeanne	8
Dennis	3
Katrina	8
Rita	2
Wilma	18

Based on this sample of seven, compute the following descriptive statistics:

- Mean, median, and mode
 - Range and standard deviation
 - Should Wilma be considered an outlier in terms of the days required to restore electric service?
 - The seven hurricanes resulted in 10 million service interruptions to customers. Do the statistics show that FP&L should consider updating its approach to emergency electric system repairs? Discuss.
41. Annual sales, in millions of dollars, for 21 pharmaceutical companies follow.

8408	1374	1872	8879	2459	11413
608	14138	6452	1850	2818	1356
10498	7478	4019	4341	739	2127
3653	5794	8305			

- Provide a five-number summary.
- Compute the lower and upper limits.
- Do the data contain any outliers?
- Johnson & Johnson's sales are the largest on the list at \$14,138 million. Suppose a data entry error (a transposition) had been made and the sales had been entered as \$41,138 million. Would the method of detecting outliers in part (c) identify this problem and allow for correction of the data entry error?
- Show a box plot.